TED University
CMPE 491 Senior Design Project I

Project Specifications Report

Name of the Project	Attention-Guided Manipulation-Aware Detection of AI-Generated and Edited Images for Improved Cross-Dataset Generalization
Project Web Page	https://tunacantungul.github.io/manipulation-aware-detection/
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1. Introduction

1.1 Description

This project tests whether guiding a detector's attention toward the manipulated region of an image improves how well it detects AI-generated and AI-edited images that differ from its training data. The project is research-oriented: its main outputs are a controlled experimental study, the software that runs it, and a demonstration tool, and its results are intended for publication as a research paper.

Problem. Detectors of AI-generated and AI-edited photographs often report near-perfect accuracy on the dataset they were trained on. Their accuracy drops sharply on images from unseen generators, unseen editing tools, unseen datasets, or after ordinary post-processing such as compression and resizing. A likely cause is shortcut learning: the detector relies on cues that happen to be correlated with the label in the training data, such as file format or global traces of the generator, instead of on the manipulation itself. Such cues disappear when the data changes, and the detector fails with them.

Research question. Can an attention mechanism added to an existing detector steer its decisions toward manipulation-relevant regions and thereby improve cross-dataset performance? We divide this into three sub-questions:

1. Does an attention module trained only with image-level labels (real or manipulated) change cross-dataset performance compared with the unmodified detector?
2. Does supervising the same module with manipulation masks improve cross-dataset performance beyond the unsupervised module?
3. Does the attention of each variant stay on the manipulated region when the data distribution changes, and is this associated with correct decisions on individual images?

Approach. We take an existing detector with public code and compare variants of it that differ only in the attention component: the unmodified detector, the detector with an attention module trained with image-level labels only, and the same detector whose attention module is also trained

to match a manipulation mask. Control variants that receive the same kind of supervision in the wrong place separate the effect of the true manipulated region from the effect of extra regularization or of object saliency.

Testbed. The main setting is photographs locally edited by generative inpainting. In these images the edited region is known by construction, so masks for training and evaluation are available without manual annotation. Fully AI-generated images and traditionally edited images (splicing, copy-move) are used as more distant test conditions.

Evaluation. Every variant is evaluated under three kinds of distribution shift named in the proposal: cross-generator (unseen editing models), cross-dataset (datasets built with different pipelines), and post-processing (JPEG compression and resizing). In addition, we test on traditionally edited images, on a shortcut probe in which the original pixels outside the edited region are restored, and, in the extended setting, on fully AI-generated images. For each condition we measure both detection performance and whether the detector's attention stays on the manipulated region.

Scope. The project covers image-level detection (manipulated or authentic) of still images. Pixel-level localization is used only to analyze where the detector looks, not as a separate product. Video, audio and face-specific deepfake detection are outside the scope. The project does not aim to build a new detector from scratch, nor a production forensic tool.

Expected outputs.

- A reproducible training and evaluation pipeline with all detector variants.
- Experimental results for each variant under each type of shift, including attention alignment measurements.
- A simple demonstration application that shows a detector's decision and where it looked in an uploaded image.
- A research paper reporting the findings, and the project reports required by the course.

1.2 Constraints

Economic. The project has no dedicated budget. All experiments must run on the team members' own computers and on free or low-cost cloud resources such as free-tier notebook services, or on university resources if they become available. This limits GPU memory, training time and storage. The methodology is therefore defined at two levels: a minimum setting that must be completed with modest hardware (one CNN-based detector, subsets of the training data, a small number of random seeds), and an extended setting (a second, transformer-based detector, full datasets, more seeds) that is carried out only if more resources become available. All datasets, pretrained models and software libraries used must be free for academic use.

Environmental. Training deep networks consumes electricity. The project limits this by reusing pretrained weights, fine-tuning instead of training from scratch, training on data subsets in the minimum setting, and avoiding unnecessary hyperparameter searches. Training runs are logged so that no experiment has to be repeated because its result was lost.

Social. Realistic AI-edited images can spread misinformation, damage reputations and reduce public trust in photographs. Detectors that work only on their own benchmark give a false sense of security. The project contributes to more reliable detection, but its outputs are research results and must not be presented as a verdict on whether a specific real-world image is genuine.

Political. Manipulated images are used in political campaigns and around elections, and regulation increasingly requires that AI-generated or manipulated content be disclosed, for example the transparency obligations in Article 50 of the EU Artificial Intelligence Act [1], which apply from 2 August 2026. Detection research is relevant to these debates, but a detector with non-zero error rates should not be used as sole evidence in politically sensitive cases. The project will report error rates clearly for this reason.

Ethical. A false positive can wrongly mark an authentic photograph, and therefore the person who took or appears in it, as fake. Research on detector weaknesses is also dual-use, because it can show manipulators how to avoid detection. The project reports false-positive rates explicitly and discusses limitations. Public datasets may contain images of people; they are used only as provided, for research, and under their licenses. These issues are discussed further in Section 1.3.

Health and safety. The project is a software and research project with no physical product, so it poses no direct health or safety risk. Some public manipulation datasets may contain disturbing content; team members may skip inspecting such images manually.

Manufacturability. For a research project, manufacturability corresponds to reproducibility: another researcher must be able to rebuild the same results. The project uses open-source code and public datasets, fixes random seeds, records all data splits, thresholds and hyperparameters, and will release the code and split lists.

Sustainability. New image generators and editing tools appear every few months, so any detector becomes outdated. The project addresses this in two ways: its research question is about generalization to unseen tools, and its pipeline is modular so that new datasets, generators and base detectors can be added without changing the rest of the system.

1.3 Professional and Ethical Issues

The team follows the ACM Code of Ethics and Professional Conduct [3], the IEEE Code of Ethics [4] and the Software Engineering Code of Ethics and Professional Practice [5]. The points below describe how these apply to this project.

Honest reporting of results. Both codes require honest and realistic claims about the work. We will report all planned experiments, including those in which attention guidance does not help, together with confidence intervals and statistical tests. Hyperparameters and decision thresholds are chosen only on in-domain validation data and never tuned on the shifted test sets, so that cross-dataset results are not inflated. Our hypotheses are written down before the experiments are run.

Avoiding harm and overclaiming. A detector's output is a probability, not proof. The demonstration application will show the score and the attention map, state that the result can be wrong, and report the known error rates. We will not describe the system as a forensic tool suitable for legal, journalistic or political decisions.

Data licensing and attribution. All datasets, pretrained models and code are used under their licenses and cited. Datasets that forbid redistribution are not re-published; we release only split lists and processing code.

Privacy. Public datasets may contain images of identifiable people. These images are used only for research and are not re-shared or used to identify anyone. Images uploaded to the demonstration application are processed in memory and are not stored. Where personal data of people in Türkiye is involved, the principles of the Personal Data Protection Law No. 6698 (KVKK) [2] apply.

Dual use. Analyses of where detectors fail could help someone evade them. We will describe weaknesses at the level needed for scientific understanding and will not publish tools designed to evade detection.

Academic integrity and teamwork. Prior work is cited wherever it is used, including code and figures. All three team members are expected to contribute; tasks and contributions are tracked in the project backlog. Because the results will also be submitted as a research paper, author contributions and the role of the supervisor will be stated transparently, and the rules of the target venue regarding prior publication will be checked before reports are made public.

1.4 Literature Review

Prior work shows that detectors generalize poorly, that part of the reason is shortcut learning, and that attention guidance has helped in related settings but has not been tested in a controlled way on AI-edited photographs. This section summarizes that evidence.

1.4.1 Generalization Failures and Shortcuts

Early work showed that a ResNet-50 trained on images of one GAN transfers to other GANs only with careful augmentation [6]. Detectors trained on GAN images later failed on diffusion images, because anything unfamiliar was classified as real [7], and a recent evaluation found that most off-the-shelf detectors label hard, realistic AI-generated images as real [8]. Partially edited photographs are harder still: when the inpainted region covers less than 5% of the image, detectors trained on fully synthetic images perform close to chance [9].

Geirhos et al. call decision rules that work on a benchmark but fail under changed conditions shortcuts [10]. Image forensics has concrete examples. In a widely used benchmark, real images were JPEG files of varying size while fake ones were fixed-size PNG files, and removing this difference raised cross-generator accuracy by more than 11 percentage points [11]. For inpainted images, restoring the original pixels outside the edited region reduced detector accuracy from above 0.91 to about 0.55, indicating that detectors used global traces of the generator's autoencoder rather than the edited content [12]. Manipulation datasets also tend to edit salient objects, so a detector may appear to find the edit while actually responding to the object [13].

1.4.2 Attention in Forensic Detectors

Several strong detectors contain attention components. MVSS-Net fuses edge and noise branches with dual attention [14], TruFor combines RGB content with a learned noise fingerprint through transformer-based fusion [15], and FatFormer adds forgery-aware adapters with image and frequency attention to a CLIP encoder, improving accuracy on unseen diffusion models by 9.6 points over a frozen-CLIP baseline [16]. In each case attention is introduced together with other design changes, so its own effect on generalization is unknown, and it is not constrained to the manipulated region. A controlled benchmark of manipulation detectors showed that backbone choices that change how spatial context is aggregated affect robustness [17], which supports isolating the attention component in an experiment.

1.4.3 Guiding Where a Model Looks

Outside forensics, several methods train a network's spatial focus directly. GAIN makes attention maps part of training and can use pixel-level supervision [18], Right for the Right Reasons penalizes input gradients outside annotated regions and generalizes better when training and test conditions differ [19], and Rao et al. compare such guidance losses systematically, introducing the Energy-based Pointing Game (EPG) as both a metric and a loss [20]. Their results show better worst-group accuracy under distribution shift but a trade-off between localization and accuracy in distribution, so guidance should be judged under shift.

In forensics, the closest precedent added a supervised attention layer to face-manipulation detectors; supervision greatly improved localization, but cross-dataset AUC gains were uneven, from 75.6% to 84.2% on one test set and from 63.9% to 64.4% on another [21]. A recent benchmark with pixel-level artifact masks for fully synthetic images found that aligning a detector's attention with the masks raised cross-dataset F1 while accuracy changed little, and that generic saliency masks helped less than artifact masks [22]. Attention and attribution maps must be validated before interpretation: Grad-CAM [23] is the standard map for detectors without an attention module, but some saliency methods barely change when the model's weights are randomized [24].

1.4.4 Datasets for Locally Edited Images

Semi-Truths provides about 1.47 million AI-edited versions of 27,600 real images with 223,400 masks and edit metadata [25]. TGIF provides about 75,000 text-guided inpainting forgeries made with several Stable Diffusion models and a commercial tool, in a spliced variant and a fully regenerated variant [26]. For traditional edits, IMDL-BenCo defines a standard cross-dataset protocol and a cleaned version of a test set with label leakage [17].

1.4.5 Research Gaps

1. The effect of attention alone on cross-dataset performance has not been isolated, because it is always introduced with other changes [14], [15], [16].
2. Supervised and unsupervised attention have rarely been compared under distribution shift for edited photographs; existing comparisons concern faces [21] or fully synthetic images [22].
3. Whether attention stays on the manipulated region on unseen data, and whether this relates to correct decisions on individual images, is rarely measured.
4. Correct-looking attention can be explained by object saliency [13] or global generator traces [12], and few studies control for these confounds.

This project addresses these gaps with the controlled comparison and evaluation described in Section 1.1 and specified in Section 2.

2. Requirements

The requirements below are initial and will be refined in the Analysis Report. Priority “Must” marks the minimum setting that the project must complete with limited hardware; “Should” marks the extended setting, carried out if resources allow.

2.1 Functional Requirements

Data preparation

ID	Requirement	Priority
FR1	The system shall load inpainted images, their authentic originals and their manipulation masks from Semi-Truths [25].	Must
FR2	The system shall split the inpainting models into a seen group used for training and a held-out group used only for testing, and keep a held-out set of images from the seen models as the in-dataset reference.	Must
FR3	The system shall store real and edited training images in the same file format and quality and match their sizes by cropping, so that format and size cannot act as shortcuts [11].	Must
FR4	The system shall crop edited training images so that the crop contains the edited region and crop the mask identically.	Must
FR5	The system shall convert each mask into a target map at the resolution of the attention map, where each cell holds the fraction covered by the mask; authentic images receive an all-zero target.	Must

Detector variants

ID	Requirement	Priority
FR6	The system shall provide a CNN base detector (ResNet-50, trained as in [6]) with its original global pooling (variant A0).	Must
FR7	The system shall provide variant A1: the same detector with a spatial attention module, based on [21] and [27], that reweights an intermediate feature map before pooling, trained with image-level labels only.	Must
FR8	The system shall provide variant A2: the same detector and module, additionally trained with an alignment loss between the attention map and the mask target, using a weighted mean squared error as in [22].	Must
FR9	The system shall provide control variant C2, identical to A2 but supervised with a mask of the same area placed where it does not overlap the true edit.	Must
FR10	The system shall provide control variant C1, identical to A2 but supervised with an object saliency map of the same image.	Should
FR11	The system shall provide a CLIP ViT-B/16 base detector as in [7] with the corresponding attention variants.	Should
FR12	The system shall support the Energy loss of [20] as an alternative alignment loss.	Should

Training

ID	Requirement	Priority
FR13	All variants shall be trained with the same data, augmentation and schedule, differing only in the attention component and its loss.	Must
FR14	Hyperparameters and the decision threshold shall be selected on an in-domain validation split only.	Must
FR15	Each variant shall be trained with at least 3 random seeds (5 in the extended setting).	Must
FR16	Training shall save checkpoints and resume after interruption, since free cloud sessions can end unexpectedly.	Must

Evaluation under distribution shift

ID	Requirement	Priority
FR17	The system shall evaluate every variant on: in-dataset images (S0), unseen inpainting models (S1, cross-generator), the TGIF dataset with spliced and regenerated images reported separately [26] (S2, cross-dataset), and traditional-edit test sets under the IMDL-BenCo protocol [17] (S3).	Must
FR18	The system shall evaluate every variant on post-processed versions of S0 and S1 with at least three JPEG quality levels and two downscaling factors, following prior robustness evaluations [6], [7], [26] so that results are comparable (S4, post-processing shift). The exact levels will be fixed in the Analysis Report.	Must
FR19	The system shall evaluate every variant on images whose original pixels are restored outside the edited region (S5, shortcut probe, following [12]).	Must
FR20	The system shall evaluate every variant on fully AI-generated images with artifact masks [22] (S6).	Should
FR21	Every test condition shall include authentic images, and results on edited images shall be reported by mask area (below 5%, 5 to 20%, 20 to 50%, above 50%).	Must
FR22	The system shall report AUROC, average precision, balanced accuracy and accuracy on real and on edited images at the fixed threshold, and the true-positive rate at 1% false-positive rate.	Must

Attention analysis

ID	Requirement	Priority
FR23	The system shall compute, for each edited test image, the EPG [20], IoU and pixel-level F1 between the attention map and the ground-truth mask.	Must
FR24	The system shall compute Grad-CAM maps [23] for all variants, so that A0 can be compared with the others using the same method.	Must
FR25	The system shall apply the model-parameter randomization test [24] before attention maps are interpreted.	Must
FR26	The system shall perform a restoration test: restore the most-attended region from the original photograph and compare the change in the detector’s score with restoring the true edited region and a random region of equal area.	Should

ID	Requirement	Priority
FR27	The system shall compare variants statistically with bootstrap confidence intervals, paired tests and correction for multiple comparisons.	Must

Demonstration application

ID	Requirement	Priority
FR28	The user shall be able to upload an image through a web interface.	Must
FR29	The application shall display the selected variant’s decision (edited or authentic) and its score.	Must
FR30	The application shall display the detector’s attention map as a heatmap overlaid on the image.	Must
FR31	The user shall be able to switch between variants (A0, A1, A2) to compare their decisions and attention maps on the same image.	Must
FR32	For sample images with known masks, the application shall also show the true edited region next to the attention map.	Should
FR33	The application shall state that results can be wrong and show the measured error rates.	Must

2.2 Non-Functional Requirements

ID	Requirement	Priority
NFR1	Reproducibility: all random seeds, data splits, thresholds and hyperparameters shall be recorded, and the code and split lists shall be released publicly.	Must
NFR2	Hardware: the minimum setting shall be trainable on a single consumer GPU or a free cloud notebook, using data subsets and a reduced crop size if necessary.	Must
NFR3	Modularity: base detectors, attention modules, losses and datasets shall be interchangeable through configuration, without code changes elsewhere.	Must
NFR4	Fair comparison: A1, A2, C1 and C2 shall have identical architectures and parameter counts.	Must
NFR5	Performance: the demonstration application shall return a result for one image within 10 seconds on a standard laptop CPU.	Should
NFR6	Privacy: images uploaded to the demonstration application shall not be stored.	Must

ID	Requirement	Priority
NFR7	Usability: the demonstration application shall be usable without installation or technical knowledge, through a web browser.	Should
NFR8	Documentation: the code shall include setup and usage instructions, and all reports shall be published on the project web page.	Must
NFR9	Licensing: only datasets, models and libraries that are free for academic use shall be used, and all shall be cited.	Must

3. References

[1] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), *Official Journal of the European Union*, OJ L, 2024/1689, 12 Jul. 2024.

[2] Law No. 6698 on the Protection of Personal Data (KVKK), *Official Gazette of the Republic of Türkiye*, no. 29677, 7 Apr. 2016.

[3] Association for Computing Machinery, "ACM Code of Ethics and Professional Conduct," 2018.

[4] IEEE, "IEEE Code of Ethics," IEEE Policies, Section 7.8.

[5] ACM/IEEE-CS Joint Task Force on Software Engineering Ethics and Professional Practices, "Software Engineering Code of Ethics and Professional Practice," 1999.

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